

Validation of cut-cell immersed boundary method for DNS application

Artur Gesla

München, 23.04.2021

Presentation plan

- Motivation of the project
- Test cases
 - Oblique pipe flow
 - Flow around a circular cylinder
- Summary & future work

Motivation

- Validate the cut-cell method implemented in MGLET.
- Immersed boundary methods implemented in MGLET:
 - Ghost-cell
 - Cut-cell
- Cut cell method
 - wall shear stress is readily available
 - mass and momentum are conserved
 - can be computationally more expensive
- Validation cases:
 - transient oblique pipe flow:
 - cartesian grid is non-conforming to the oblique pipe geometry
 - transient behaviour can be evaluated
 - flow around circular cylinder
 - flow separations can be evaluated

MGLET

- **M**ulti **G**rid **L**arge **E**ddy **T**urbulence
- solves incompressible Navier-Stokes equations for (u,v,w) velocities and pressure
- performs DNS and LES simulations
- Spatial discretisation
 - employs 2nd order finite volume method
 - uses cartesian grids
 - staggered grid arrangement
 - handles arbitrarily shaped boundaries by ghost-cell and cut-cell immersed boundary methods
- Spatial integration
 - Gauss-Seidel on fine grids
 - Strongly Implicit Procedure on coarsest grid
- Time integration
 - low-storage 3rd order Runge-Kutta
- Chorin's projection method to ensure divergence free velocity field

Immersed boundary methods

Ghost-cell

- cells inside and outside the solid body are used for the interpolation at the solid boundary

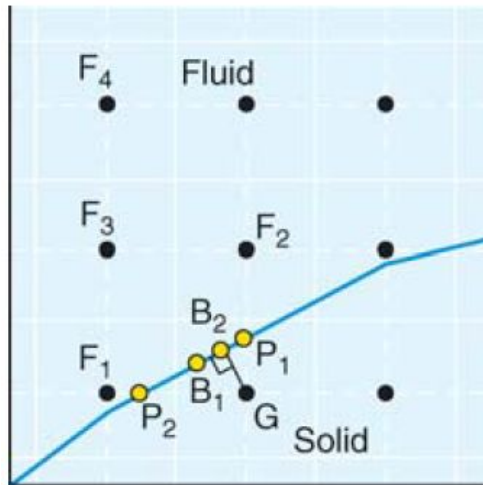


Fig. 1: Ghost-cell illustration [1]

Cut-cell

- solid boundary cuts the cartesian cells and forms new finite volume cells
- flux integrals are evaluated at cell faces for discretisation

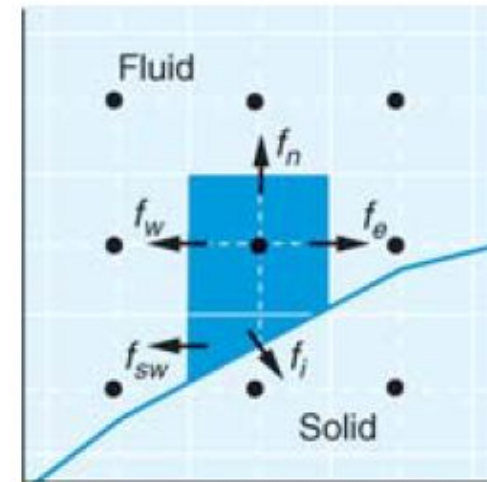
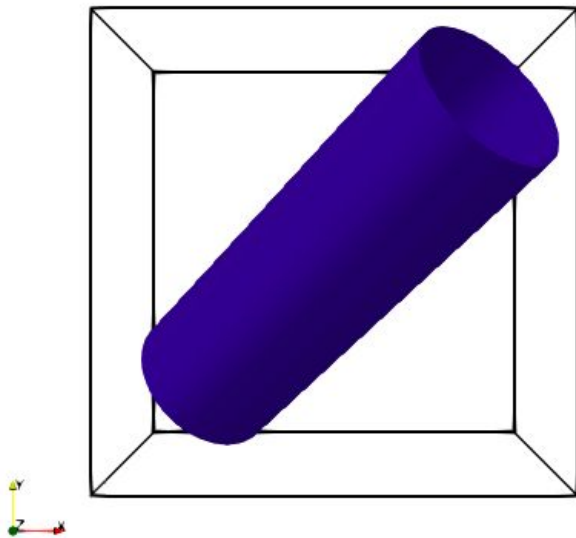


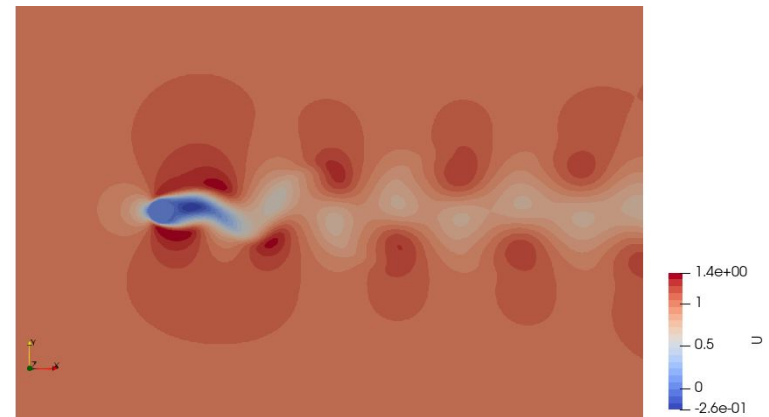
Fig. 2: Cut-cell illustration [1]

Validation cases

- Transient oblique pipe flow:
 - diverse cut geometries
 - grid convergence study
 - temporal convergence study
 - wall shear stress convergence study

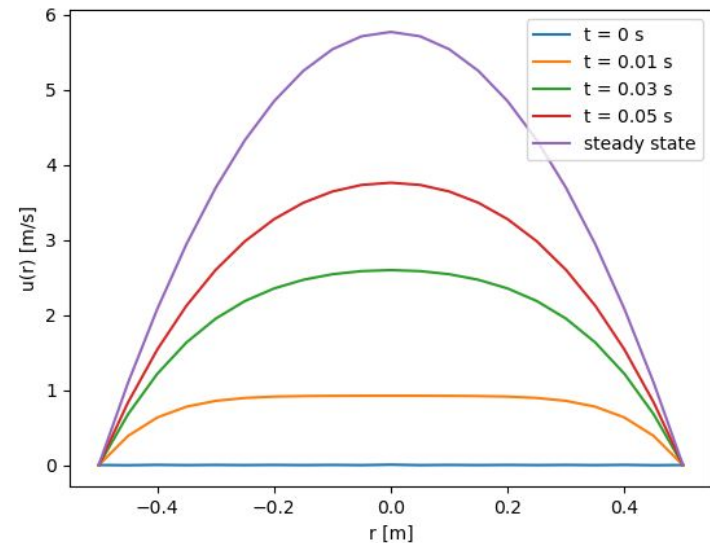
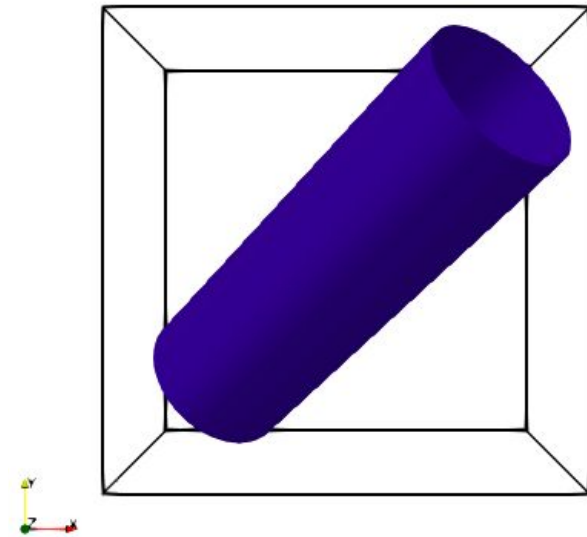


- Flow around a circular cylinder:
 - flow separation
 - lift coefficient evaluation
 - drag coefficient evaluation
 - vortex shedding frequency evaluation



Transient oblique pipe flow

- Flow is driven by the pressure gradient
- Fluid is initially at rest
- There exists an analytical solution [2]
- Validation is performed for the $t = 0.01$ s
- Geometry:
 - Pipe axis: $[1,1,1]$
 - diameter = 1 m
 - STL triangles resolution: $7.5E-3$ m
 - domain $[0, 2] \times [0, 2] \times [0, 2]$ m
- Physical parameters:
 - pressure gradient in $x = -160$ Pa
 - density = 1 kg/m^3
 - dynamic viscosity = 1 kg/ms



[2] Piotr Szymanski, Quelques solutions exactes des équations d'hydrodynamique du fluide visqueux dans le cas d'un tube cylindrique, 1932

Grid study

Checks the convergence for subsequently finer grids.

Study based on:

- bulk velocity u_{bulk}
- maximal velocity u_{axial}

5 grid resolutions (in cells per diameter):

- 8 cpd, $dx=0.125$, 4096 cells
- 16 cpd, $dx=6.25E-2$, 32K cells
- 32 cpd, $dx=3.125E-2$, 262K cells
- 64 cpd, $dx=1.5625E-2$, 2.1M cells
- 128 cpd, $dx=7.1825E-3$, 16.8M cells

We expect second order convergence in space.

Time step size:

- $DT= 1E-5$
- # timesteps = 1E3
- total time of the simulation= 0.01 s
- max CFL= $6.82E-4$
- max diffusion number = 0.49

Pressure solver settings:

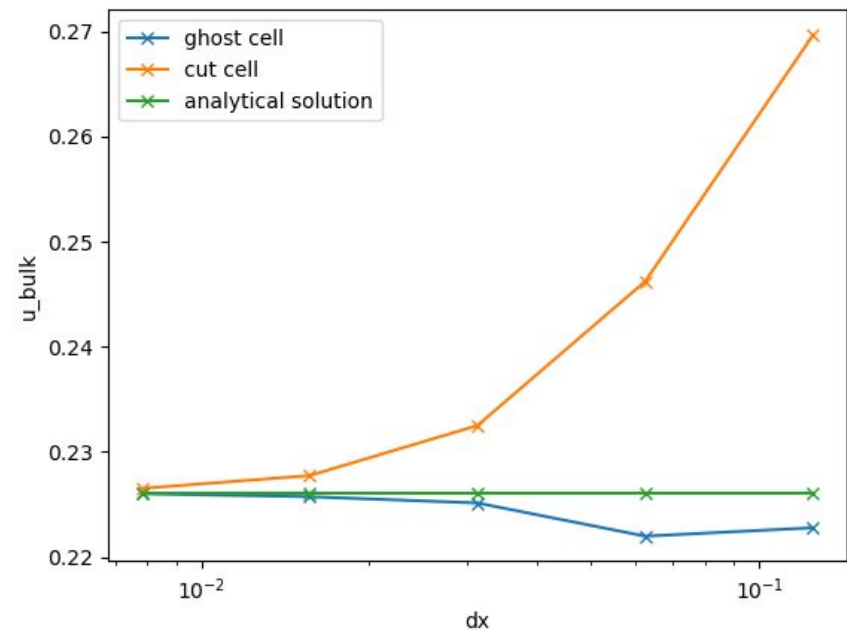
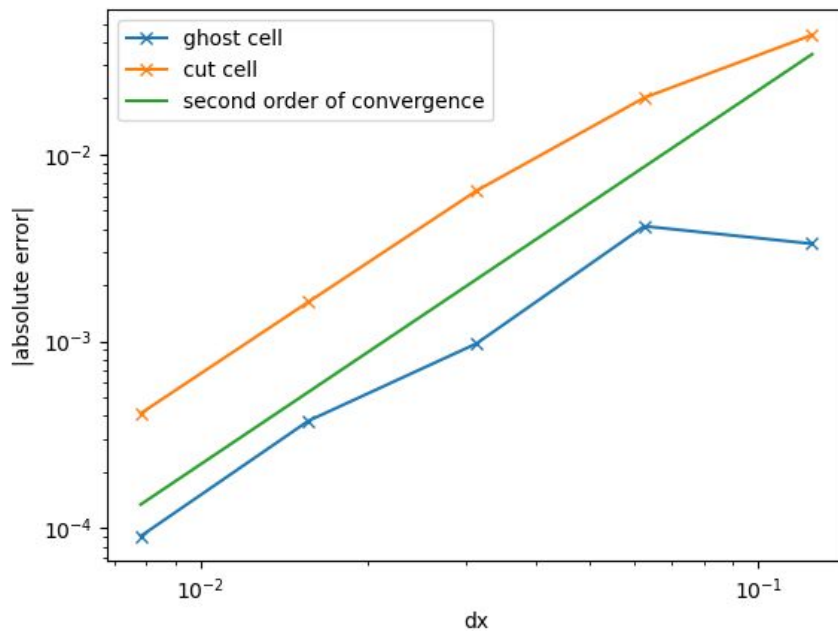
- EPCORR= $1E-6$
- MPCORR= 32
- MPCNACH= 4
- Total maximal number of pressure iterations= 128

Grid convergence study based on u_{bulk}

Study of volume averaged velocity u_{bulk} on 5 grids: 8...128 cpd

Orders of convergence ghost cell: -0.31 2.09 1.37 2.05

Orders of convergence cut cell: 1.11 1.65 1.97 1.98



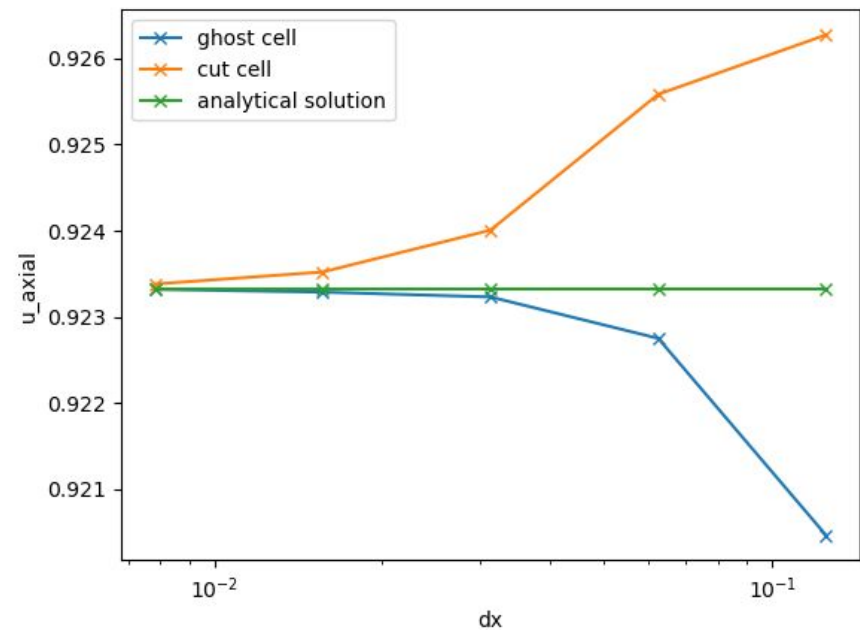
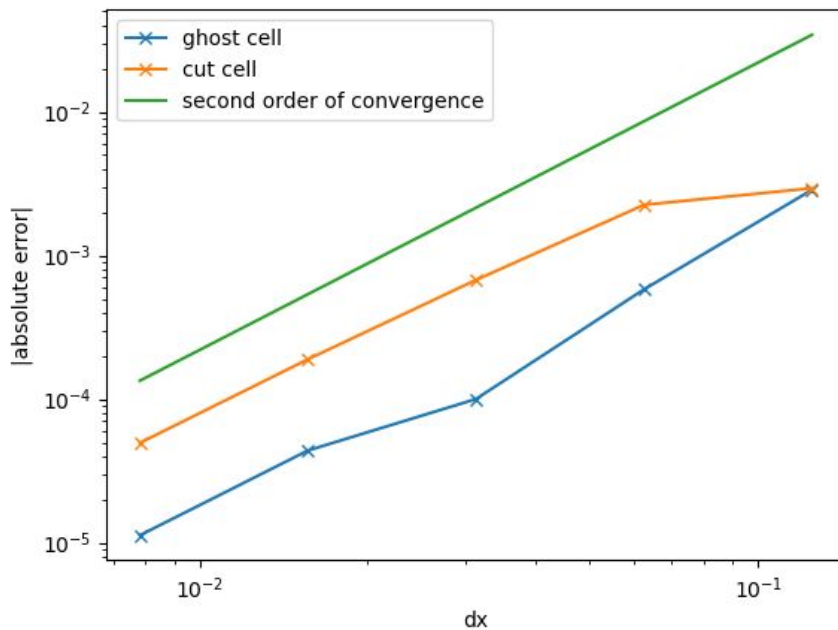
- 1) Bigger error for cut cell method
- 2) Solutions from gc and cc approach the analytical solution from opposite sides

Grid convergence study based on u_{axial}

Study of velocity u_{axial} at the point $[1,1,1]$ on 5 grids: 8...128 cpd

Orders of convergence ghost cell: 2.3 2.54 1.19 1.96

Orders of convergence cut cell: 0.39 1.74 1.83 1.93



- 1) Bigger error for cut cell method
- 2) Solutions from gc and cc approach the analytical solution from opposite sides

Oblique pipe - temporal convergence study

Checks the convergence for subsequently smaller time steps.

Time step size:

- $DT = 4E-5$, # timesteps = 250
- $DT = 2E-5$, # timesteps = 500
- $DT = 1E-5$, # timesteps = 1000
- $DT = 5E-6$, # timesteps = 2000
- $DT = 2.5E-6$, # timesteps = 4000

Total time of the simulation: 0.01 s

We expect third order convergence in time.

Pressure equation settings:

- $EPCORR = 1E-11$
- $MPCORR = 32$
- $MPCNACH = 30$
- Total maximal number of pressure iterations = 960

Grid resolution:

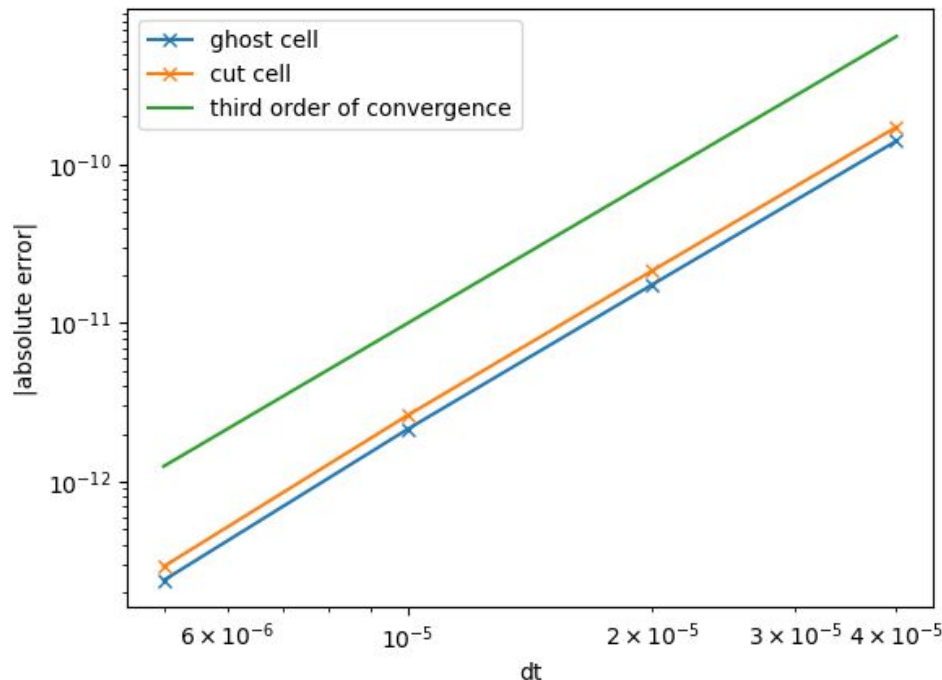
- 32 cpd, $dx = 3.125E-2$, 262K cells

Time step convergence study based on u_{bulk}

Study of volume averaged velocity u_{bulk} , 5 time steps: $4\text{E-}5 \dots 2.5\text{E-}6$

Orders of convergence ghost cell: 3.01 3.0 3.0

Orders of convergence cut cell: 3.01 3.0 3.0



- 1) Orders of convergence are based on three subsequent solutions.
- 2) For plotting it was assumed that the solution for the smallest dt is the exact solution.
- 3) u_{bulk} at smallest dt :
 - a) analytical solution = 0.2261
 - b) ghost cell = 0.2251
 - c) cut cell = 0.2325

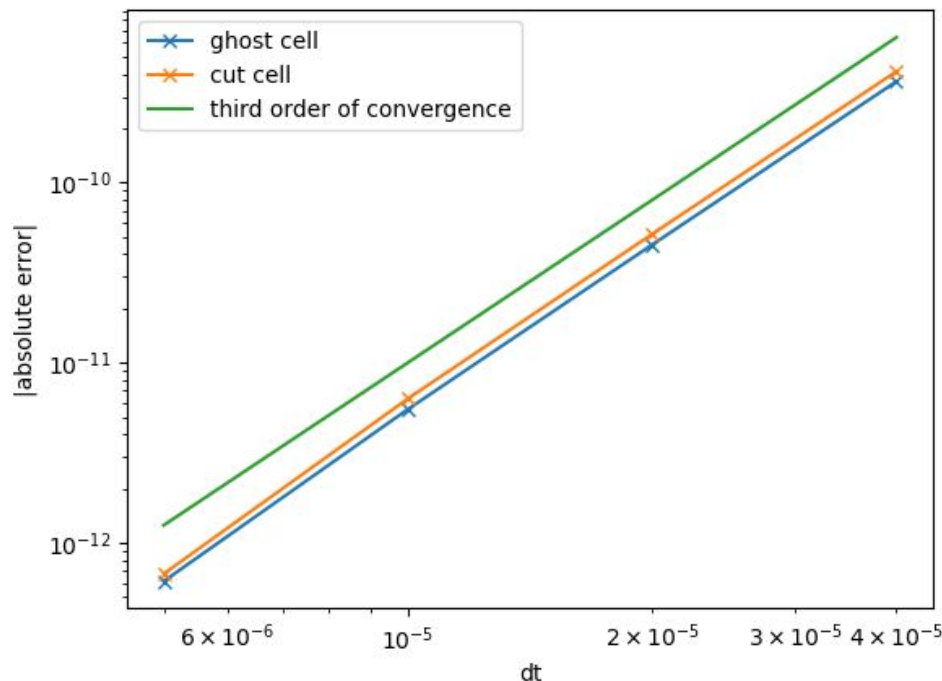
Spatial error is the dominant contribution to the overall error.

Time step convergence study based on u_{axial}

Study of velocity u_{axial} at the point $[1,1,1]$, 5 time steps: $4\text{E-}5 \dots 2.5\text{E-}6$

Orders of convergence ghost cell: 3.01 3.02 3.0

Orders of convergence cut cell: 3.0 3.01 3.07



- 1) Orders of convergence are based on three subsequent solutions.
- 2) For plotting it was assumed that the solution for the smallest dt is the exact solution.
- 3) u_{axial} at smallest dt :
 - a) analytical solution = 0.9233
 - b) ghost cell = 0.9232
 - c) cut cell = 0.9240

Spatial error is the dominant contribution to the overall error.

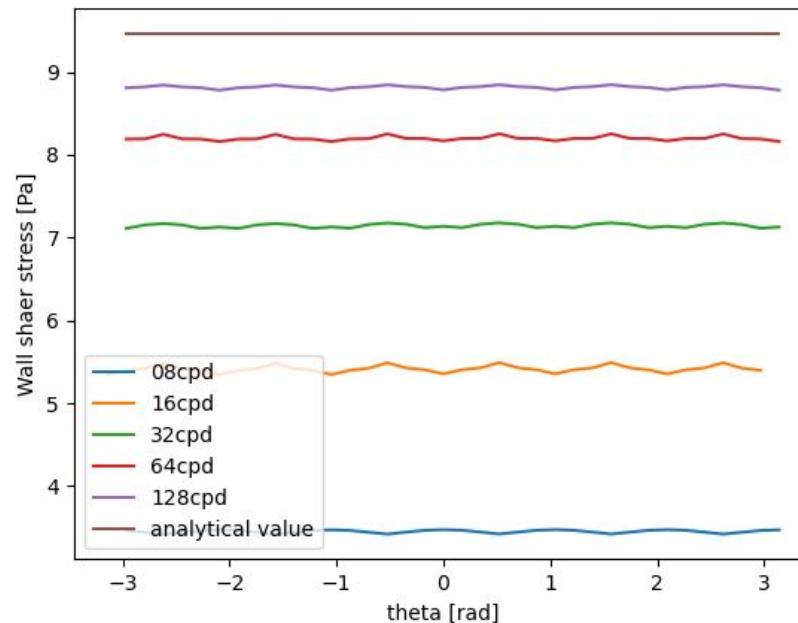
Wall shear stress in ghost cell

In the midspan of the pipe; wall shear stress is computed a posteriori based on the velocity 2 cells away from the wall. Analytical value: 9.46 Pa.

Cases: ['08cpd', '16cpd', '32cpd', '64cpd', '128cpd']

Mean values: [3.45 5.41 7.14 8.2 8.81]

Coefficient of variation: [0.0052 0.0076 0.0032 0.0031 0.0021]



Wall shear stress in ghost cell

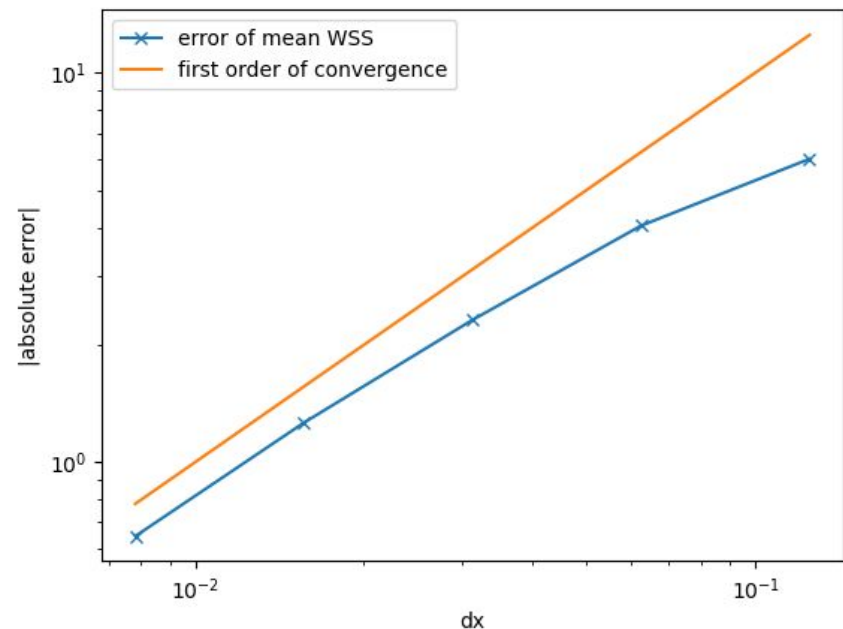
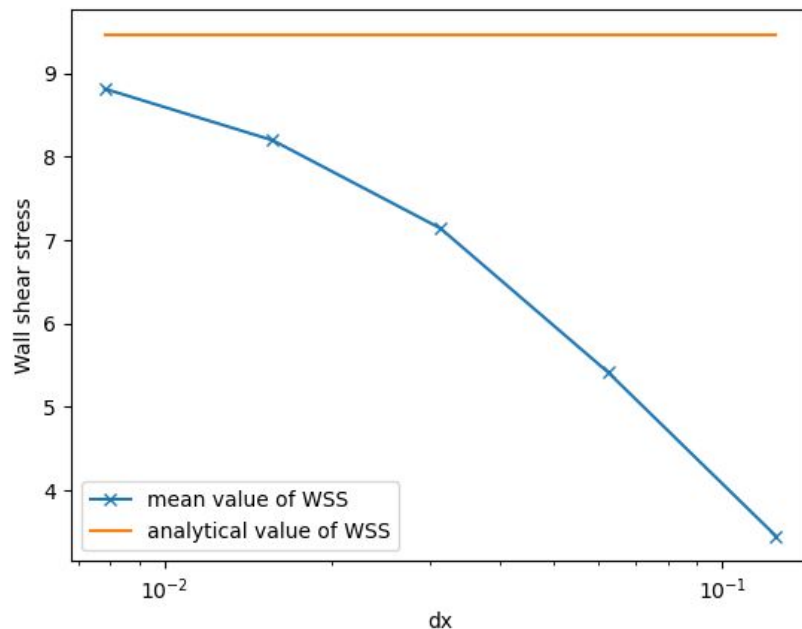
In the midspan of the pipe; wall shear stress is computed a posteriori based on the velocity 2 cells away from the wall. Analytical value: 9.46 Pa.

Cases: ['08cpd', '16cpd', '32cpd', '64cpd', '128cpd']

Mean values: [3.45 5.41 7.14 8.2 8.81]

Coefficient of variation: [0.0052 0.0076 0.0032 0.0031 0.0021]

Order of convergence: [0.57 0.8 0.88 0.97]



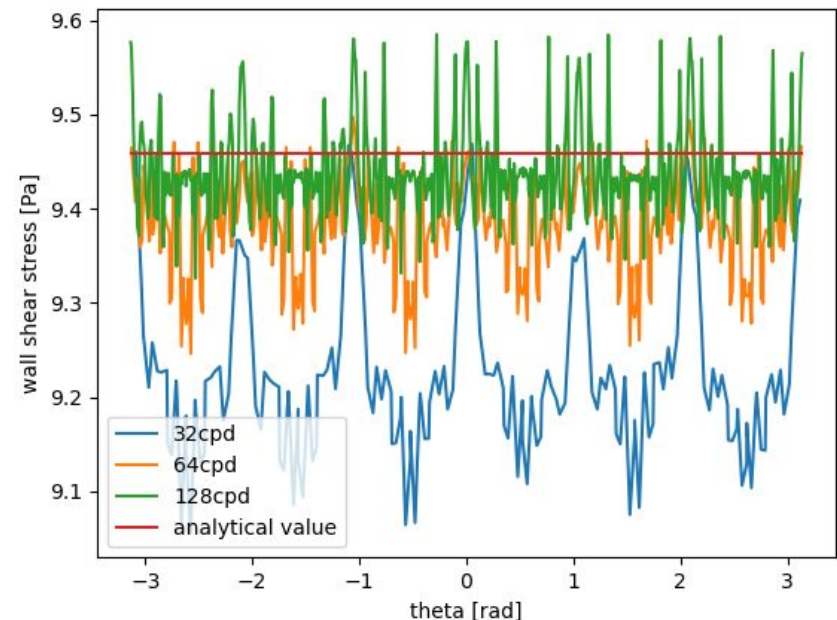
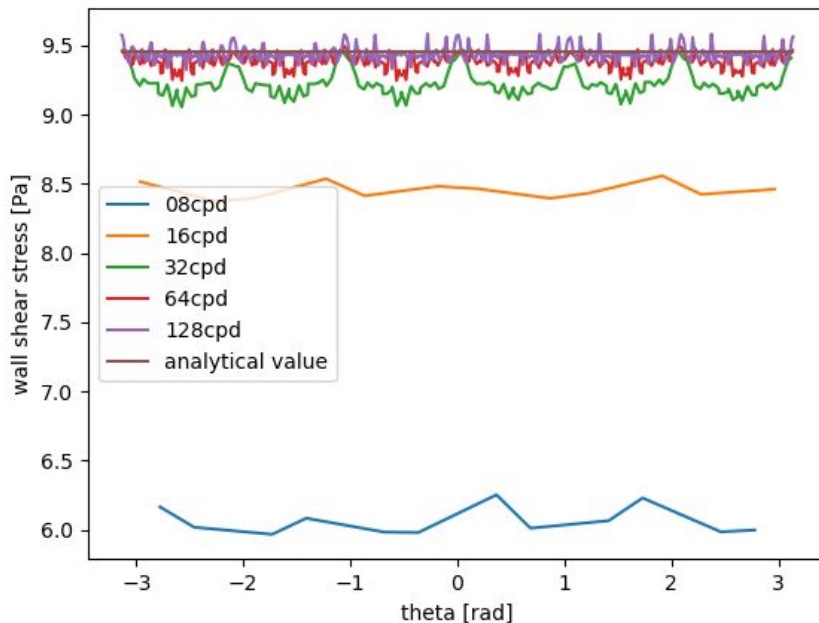
Wall shear stress in cut cell

In the midspan of the pipe; extracted directly from the solution. Analytical value: 9.46 Pa.

Cases: ['08cpd', '16cpd', '32cpd', '64cpd', '128cpd']

Mean values: [6.06 8.45 9.24 9.39 9.44]

Coefficient of variation: [0.0159 0.0067 0.011 0.0058 0.0053]



Wall shear stress in cut cell

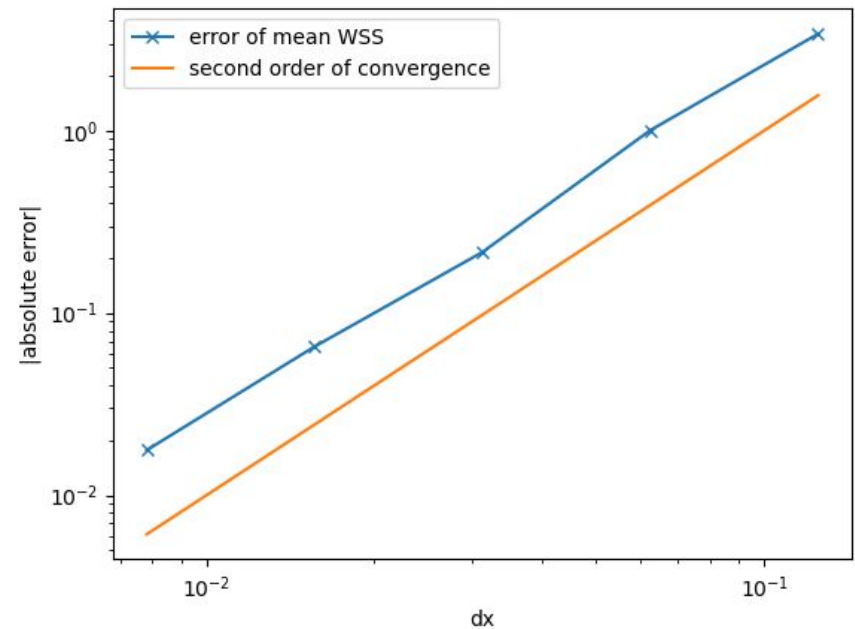
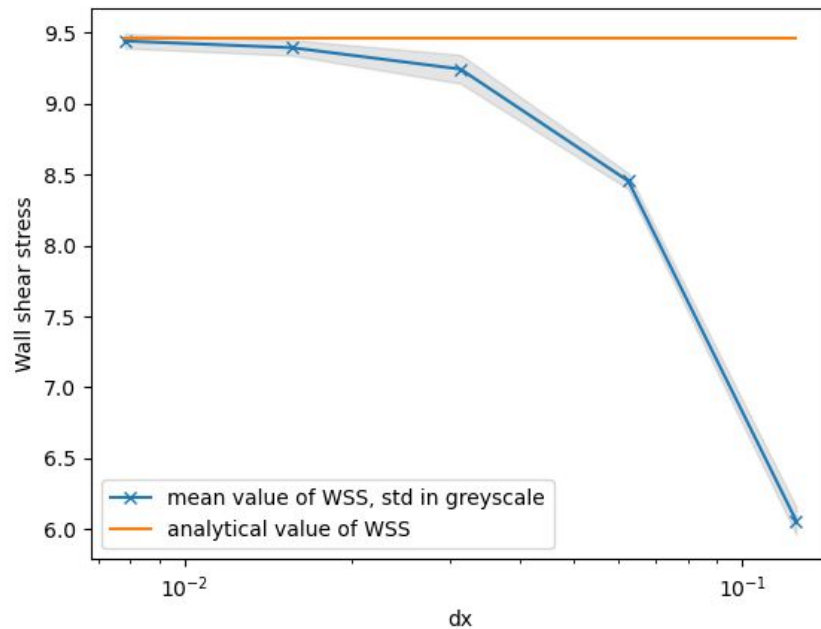
In the midspan of the pipe; extracted directly from the solution. Analytical value: 9.46 Pa.

Cases: ['08cpd', '16cpd', '32cpd', '64cpd', '128cpd']

Mean values: [6.06 8.45 9.24 9.39 9.44]

Coefficient of variation: [0.0159 0.0067 0.011 0.0058 0.0053]

Order of convergence: [1.76 2.22 1.72 1.89]



Transient oblique pipe flow

Findings:

- ✓ second order on convergence in space
- ✓ third order of convergence in time
- ✓ wall shear stress converges with second order

Flow around a circular cylinder

- Case is based on [3]
- Flow is 2D (there are 3 cells in z direction)
- Ghost-cell and cut-cell simulations
- Domain:
 - square domain: 8 m x 8 m
(26.7 D x 26.7 D)
 - cylinder diameter $D = 0.3$ m
 - grid: 1024 x 1024 elements
 - grid resolution: $dx = 7.8125E-3$
($D/dx=38.4$)

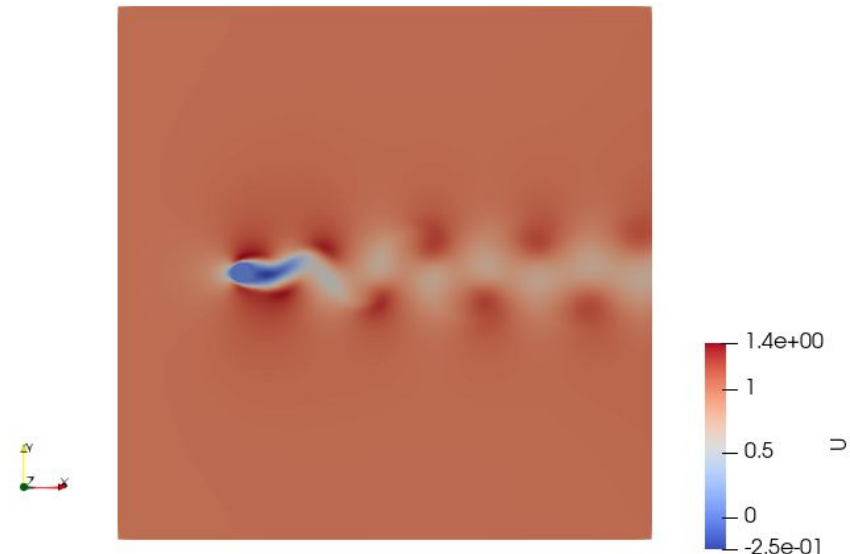


Fig. 1: U velocity at $t = 48$ s

Flow around a circular cylinder

- Boundary conditions:
 - fixed velocity $U=1$ m/s at front, top and bottom boundaries
 - zero pressure gradient at the back boundary
- Initial condition - uniform velocity $U=1$ m/s
- Physical parameters:
 - density = 1 kg/m^3
 - dynamic viscosity = $3\text{E-}3 \text{ kg/ms}$

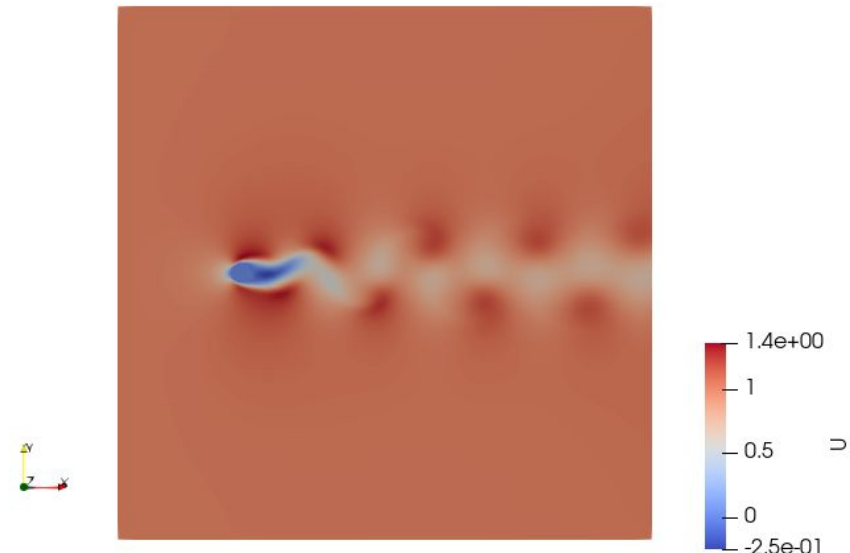


Fig. 1: U velocity at $t = 48$ s

Flow around a circular cylinder

- Time advancement settings
 - $\Delta t = 3\text{E-}3 \text{ s}$
 - # timesteps = 16 000
 - total simulated time = 48 s
- Pressure solver settings
 - EPCORR = $1\text{E-}5$
 - total number of pressure iterations = 128
- Resulting maximal CFL is around 0.6
- Resulting $\text{Re}_D = 100$

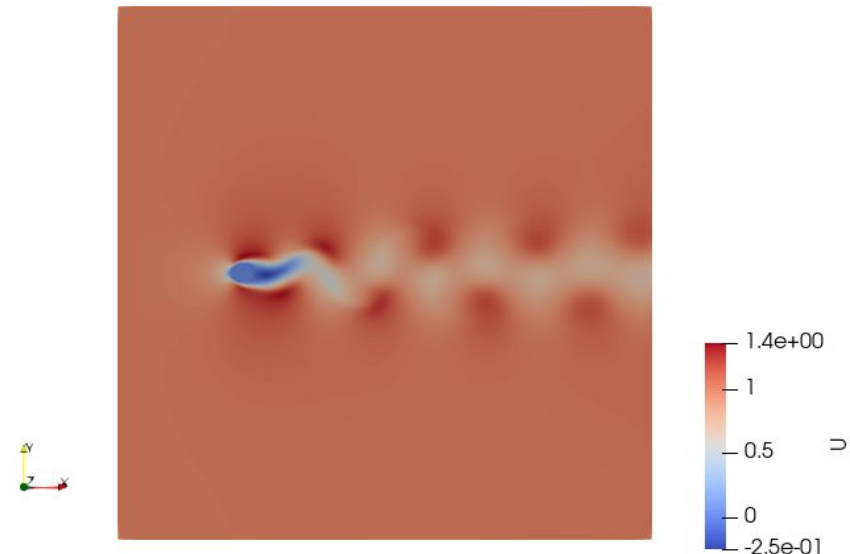


Fig. 1: U velocity at $t = 48 \text{ s}$

Comparison with literature

	Avg(Cd)	Cd'	Cl'	St
Uhlmann [3] (26.7D x 26.7D)	1.501	± 0.011	± 0.349	0.172
Uhlmann [3] (40D x 40D)	1.453	± 0.011	± 0.339	0.169
Liu et al. [4]	1.35	± 0.012	± 0.339	0.165
Present: cut-cell	1.45	± 0.012	± 0.374	0.174
Present: ghost-cell	1.40	± 0.01	± 0.347	0.173

Where:

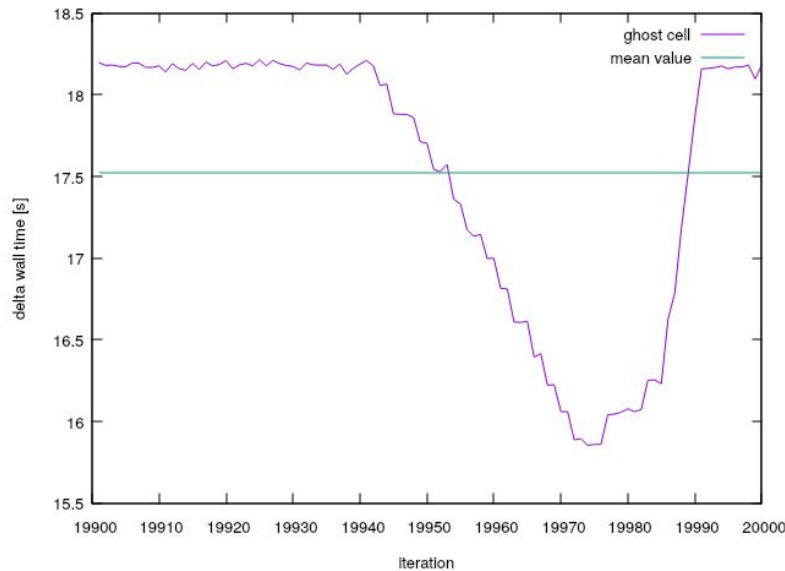
- Avg(Cd) - average value of drag coefficient Cd in steady state
- Cd' - amplitude of oscillation of drag coefficient
- Cl' - amplitude of oscillation of lift coefficient
- St - Strouhal number: $St = Df/U$

[3] Markus Uhlmann, An immersed boundary method with direct forcing for the simulation of particulate flows, Journal of Computational Physics, Volume 209, Issue 2, 2005

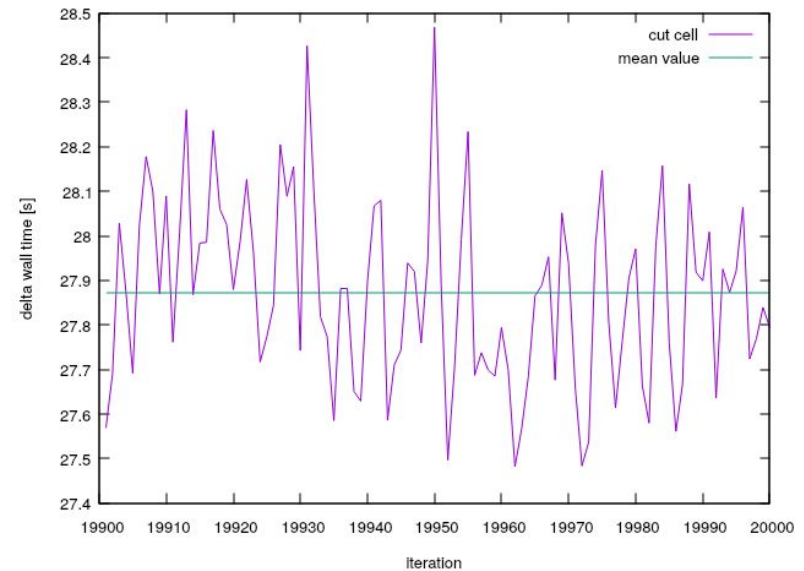
[4] C. Liu, X. Zheng, C. Sung, Preconditioned multigrid methods for unsteady incompressible flows, J. Comput. Phys. 139 (1998) 35– 57.

Performance of ghost-cell and cut-cell methods

Evaluation of performance is based on last 100 iterations from each simulation. Number of pressure iterations remained constant and equal 128.



Ghost-cell mean wall time per time step: 17.52 s



Cut-cell mean wall time per time step: 27.87 s

Summary and future work

Studies performed:

- ✓ transient oblique pipe flow:
 - grid convergence study
 - temporal convergence study
 - wall shear stress convergence study
- ✓ 2D flow around a circular cylinder
 - lift coefficient evaluation
 - drag coefficient evaluation
 - vortex shedding frequency evaluation

Findings:

- GC and CC converge in the expected order
- GC is slightly more accurate than CC
- CC is computationally more expensive
- GC and CC capture separation correctly
- whenever WSS is needed CC is recommended

Future work:

- 3D flow more complex than an oblique pipe flow (e.g. 3D flow around the cylinder from [5])
- turbulent flow case

[5] Schäfer M., Turek S., Durst F., Krause E., Rannacher R. (1996) Benchmark Computations of Laminar Flow Around a Cylinder. In: Hirschel E.H. (eds) Flow Simulation with High-Performance Computers II. Notes on Numerical Fluid Mechanics (NNFM), vol 48. Vieweg+Teubner Verlag.

Appendix

Analytical solution

Equation being solved:

$$\frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right)$$

Boundary conditions:

- No slip condition at the walls

Initial conditions:

- Fluid is initially at rest

Flow is driven by a sudden application of the pressure gradient: $G = -dp/dx$

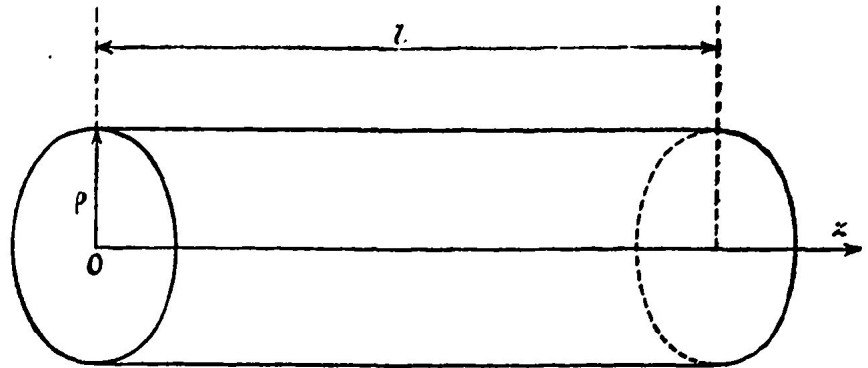


Fig. 1: Pipe [1]

Analytical solution

$$u(r, t) = \underbrace{\frac{G}{4\mu} (R^2 - r^2)}_{\text{steady}} - \underbrace{\frac{2GR^2}{\mu} \sum_1^{\infty} \frac{1}{\lambda_n^3} \frac{J_0(\lambda_n \frac{r}{R})}{J_1(\lambda_n)}}_{\text{transient}} \exp\left(-\lambda_n^2 \frac{\nu t}{R^2}\right)$$

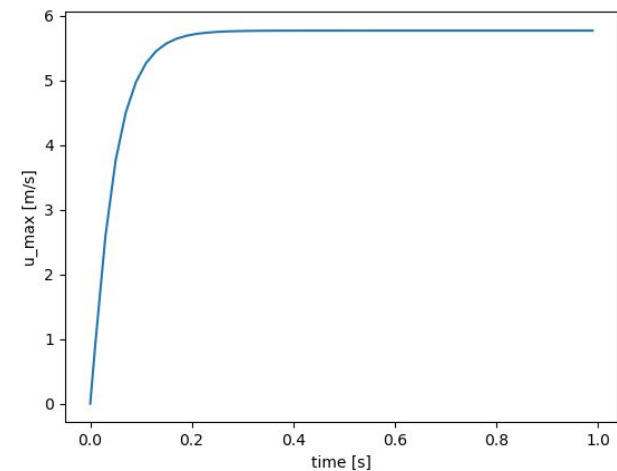
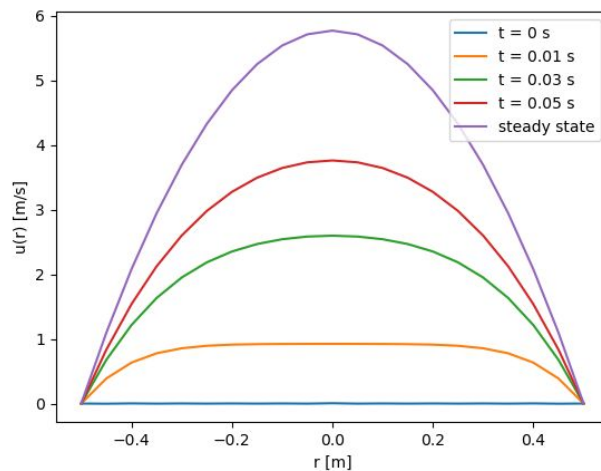
Where:

G - pressure gradient

J_0 - Bessel function of first kind and order 0

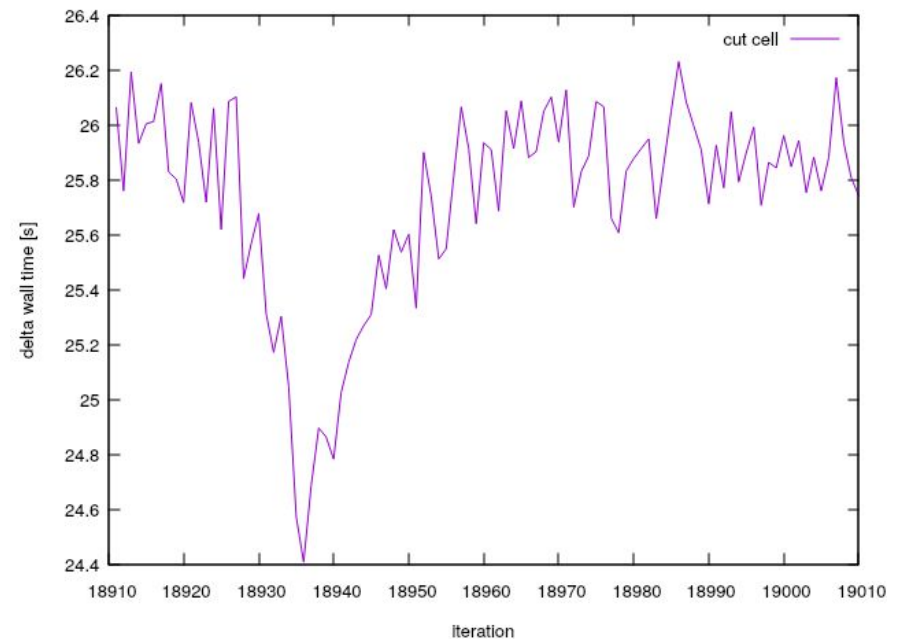
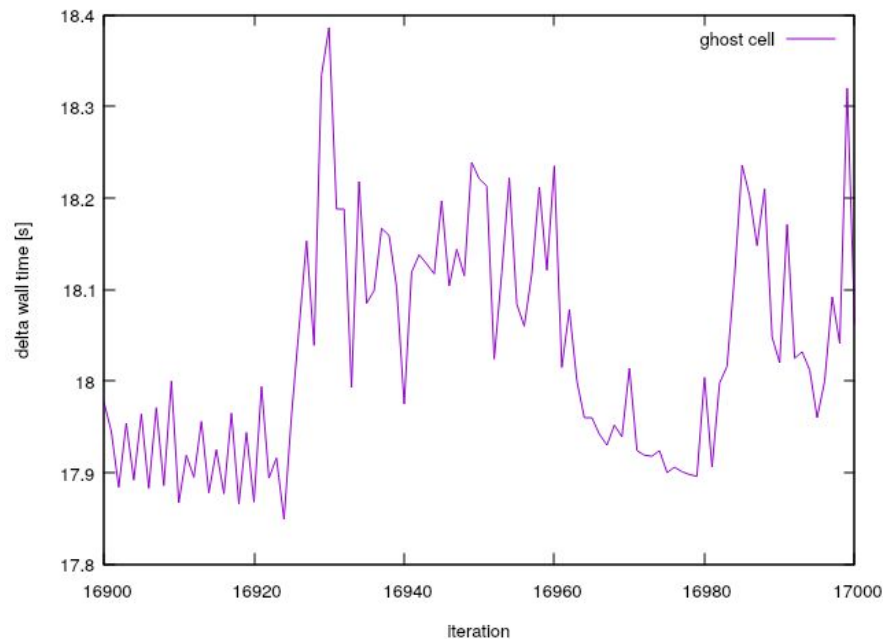
J_1 - Bessel function of first kind and order 1

lambda_n - zeros of J_0



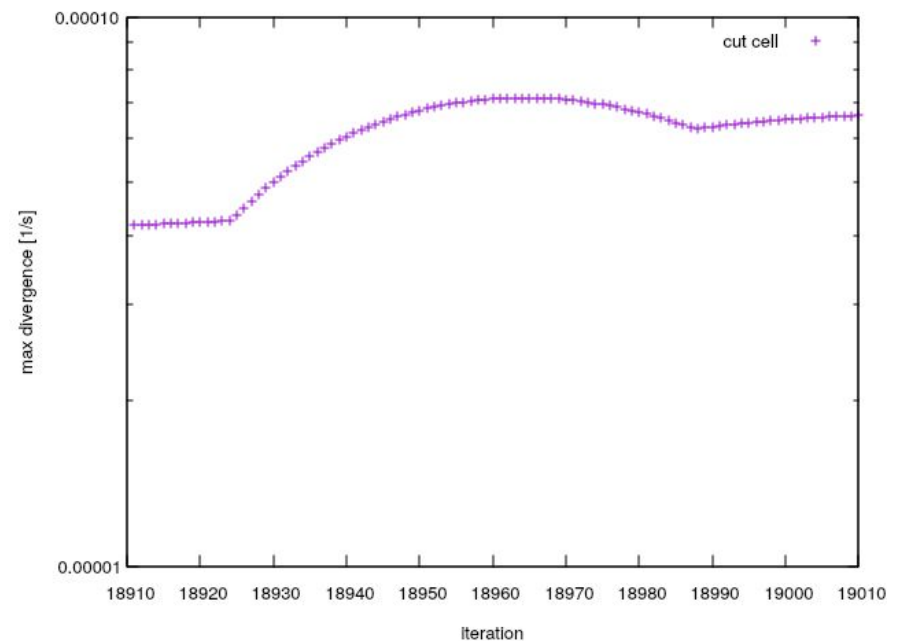
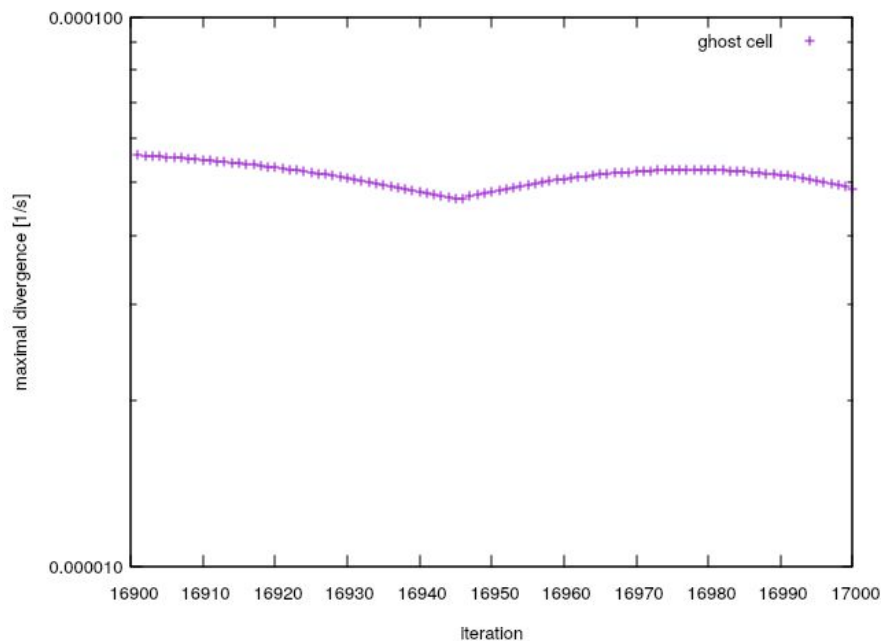
Performance of ghost-cell and cut-cell methods

Evaluation of performance is based on last 100 iterations from each simulation. Number of pressure iterations remained constant and equal 128.



Performance of ghost-cell and cut-cell methods

Evaluation of performance is based on last 100 iterations from each simulation. Number of pressure iterations remained constant and equal 128. Divergence threshold $1\text{E-}5$ was never achieved.



Old slides

Comparison with literature

	Avg(Cd)	Cd'	Cl'	St
Uhlmann [3]	1.501	± 0.011	± 0.349	0.172
Uhlmann [3] enlarged domain	1.453	± 0.011	± 0.339	0.169
Liu et al. [4]	1.35	± 0.012	± 0.339	0.165
Present	1.45	± 0.012	± 0.374	0.173

Where:

- Avg(Cd) - average value of drag coefficient Cd in steady state
- Cd' - amplitude of oscillation of drag coefficient
- Cl' - amplitude of oscillation of lift coefficient
- St - Strouhal number: $St = Df/U$

[3] Markus Uhlmann, An immersed boundary method with direct forcing for the simulation of particulate flows, Journal of Computational Physics, Volume 209, Issue 2, 2005

[4] C. Liu, X. Zheng, C. Sung, Preconditioned multigrid methods for unsteady incompressible flows, J. Comput. Phys. 139 (1998) 35– 57.

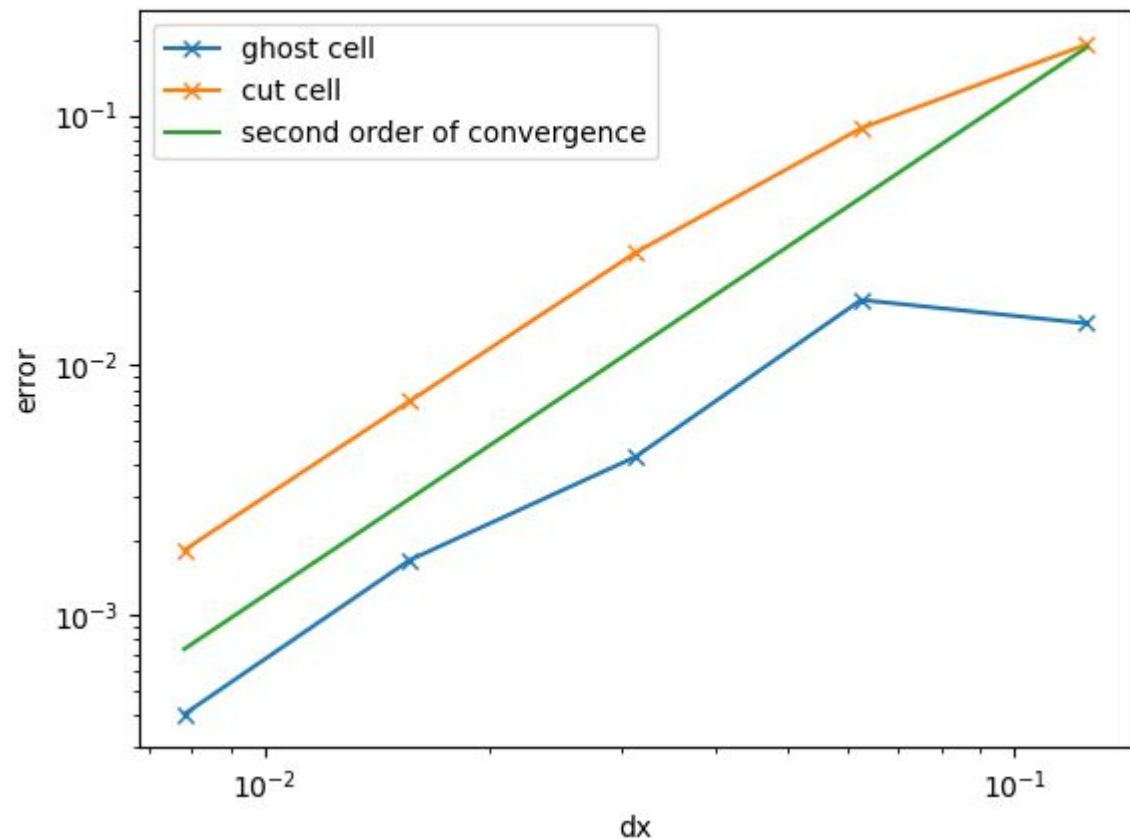
Grid convergence study

5 grids:

- 8...128 cpd

Convergence order for the subsequent refinements:

- Ghost cell
 - -0.31
 - 2.09
 - 1.37
 - 2.05
- Cut cell
 - 1.11
 - 1.65
 - 1.97
 - 1.98



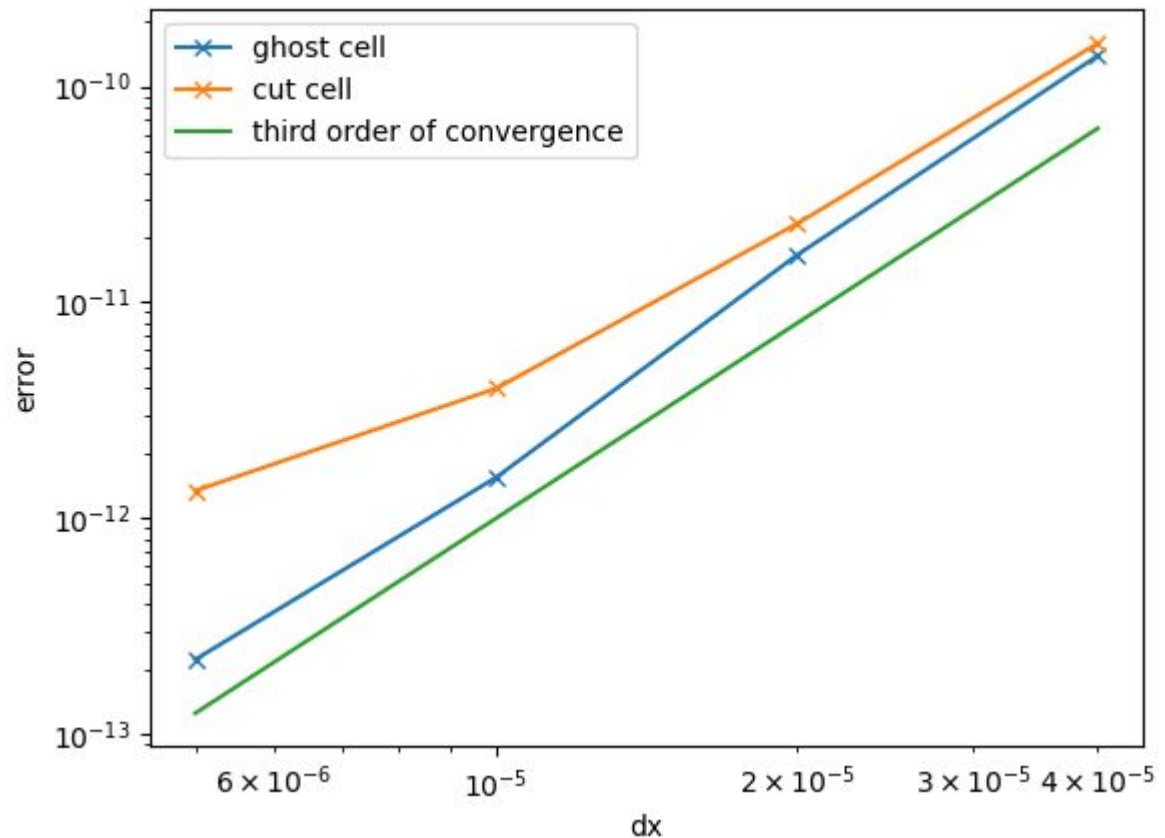
Time step convergence study

5 time steps:

- $4\text{E-}5 \dots 2.5\text{E-}6$

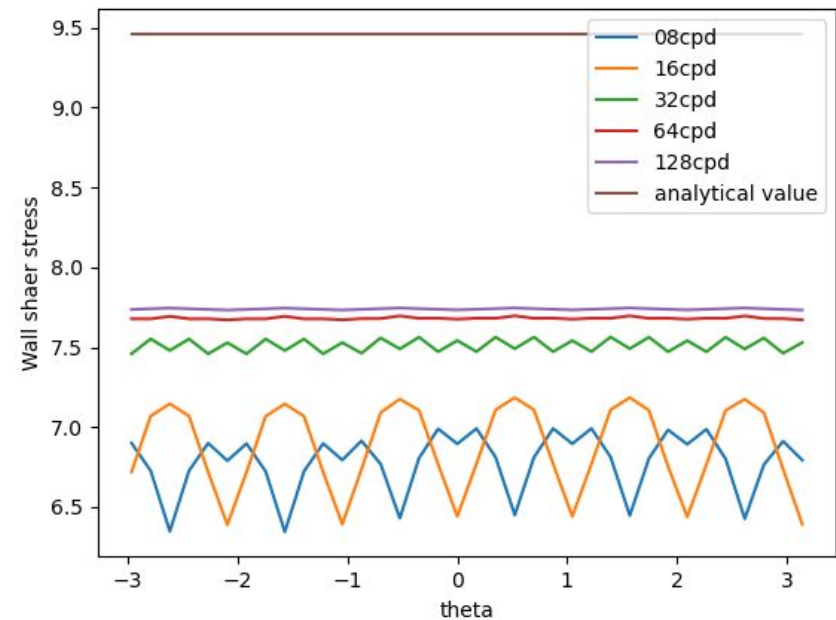
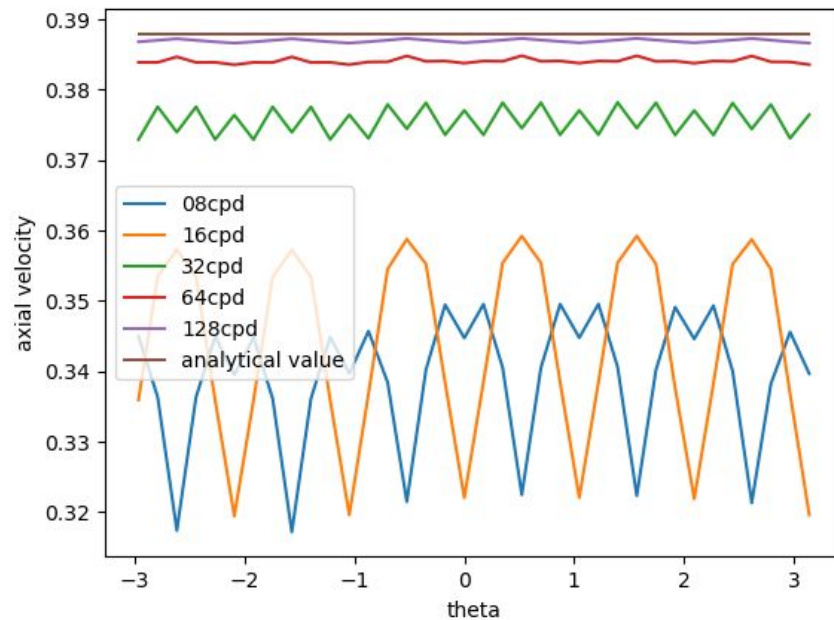
Orders of convergence
based on 3 subsequent
solutions:

- Ghost cell
 - 3.03
 - 3.50
 - 2.58
- Cut cell
 - 2.81
 - 2.85
 - 0.98



Wall shear stress in ghost cell

Calculation is based on the point 0.05 from the wall.

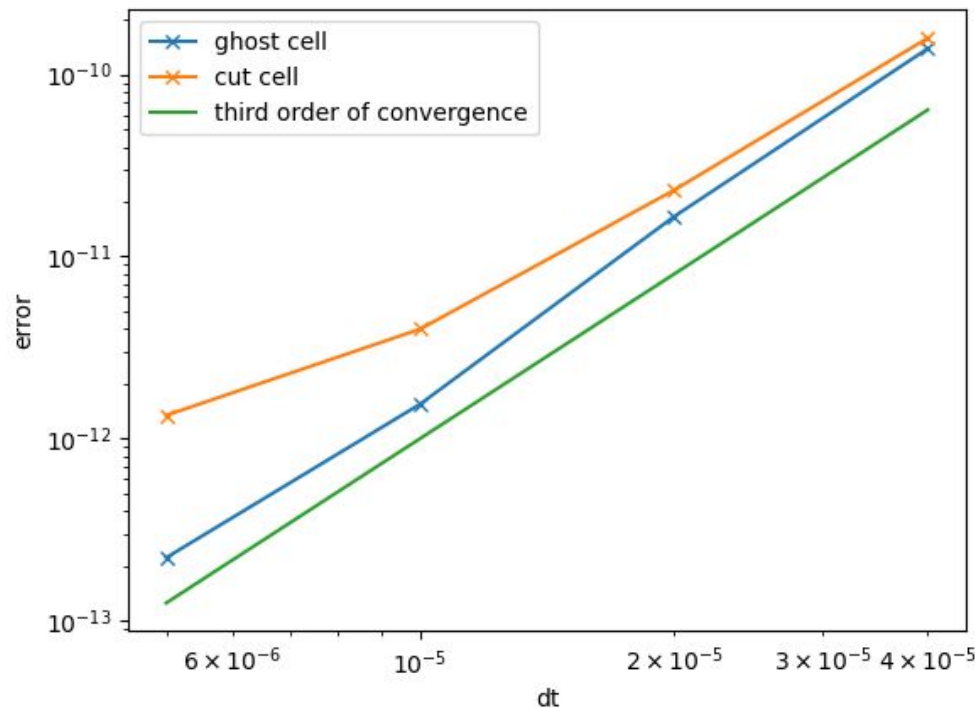


Time step convergence study based on u_{bulk}

Study of volume averaged velocity u_{bulk} , 5 time steps: $4\text{E-}5 \dots 2.5\text{E-}6$

Orders of convergence ghost cell: 3.03 3.5 2.58

Orders of convergence cut cell: 2.82 2.85 0.98



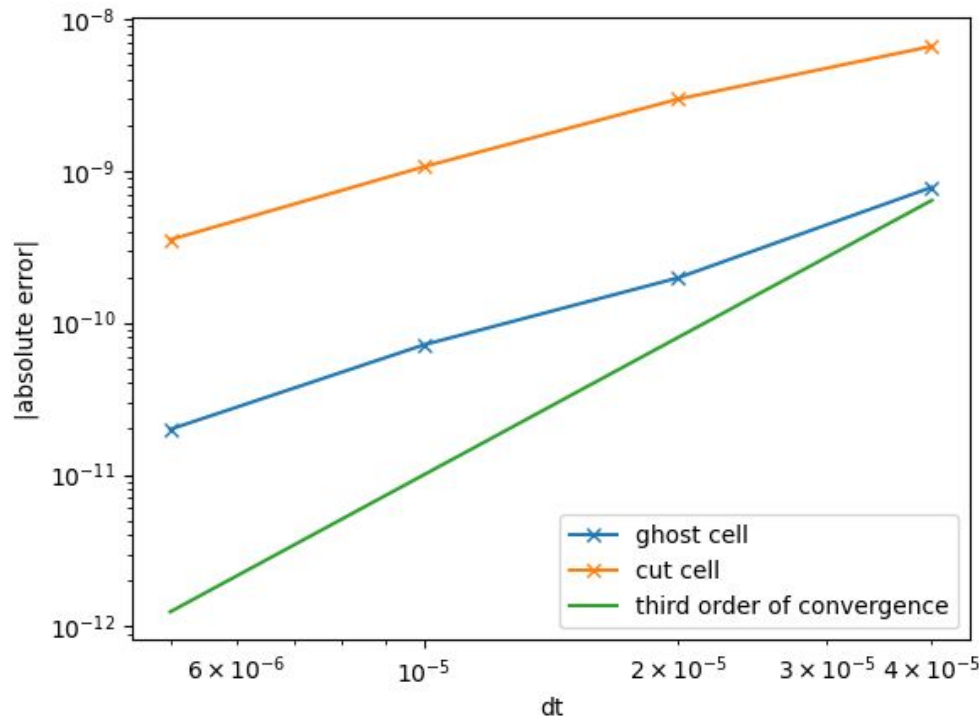
- 1) Orders of convergence are based on three subsequent solutions.
- 2) For plotting it was assumed that the solution for the smallest dt is the exact solution.
- 3) u_{bulk} at smallest dt :
 - a) analytical solution = 0.2261
 - b) ghost cell = 0.2257
 - c) cut cell = 0.2278

Time step convergence study based on u_{axial}

Study of velocity u_{axial} at the point $[1,1,1]$, 5 time steps: $4\text{E-}5 \dots 2.5\text{E-}6$

Orders of convergence ghost cell: 2.21 1.28 1.38

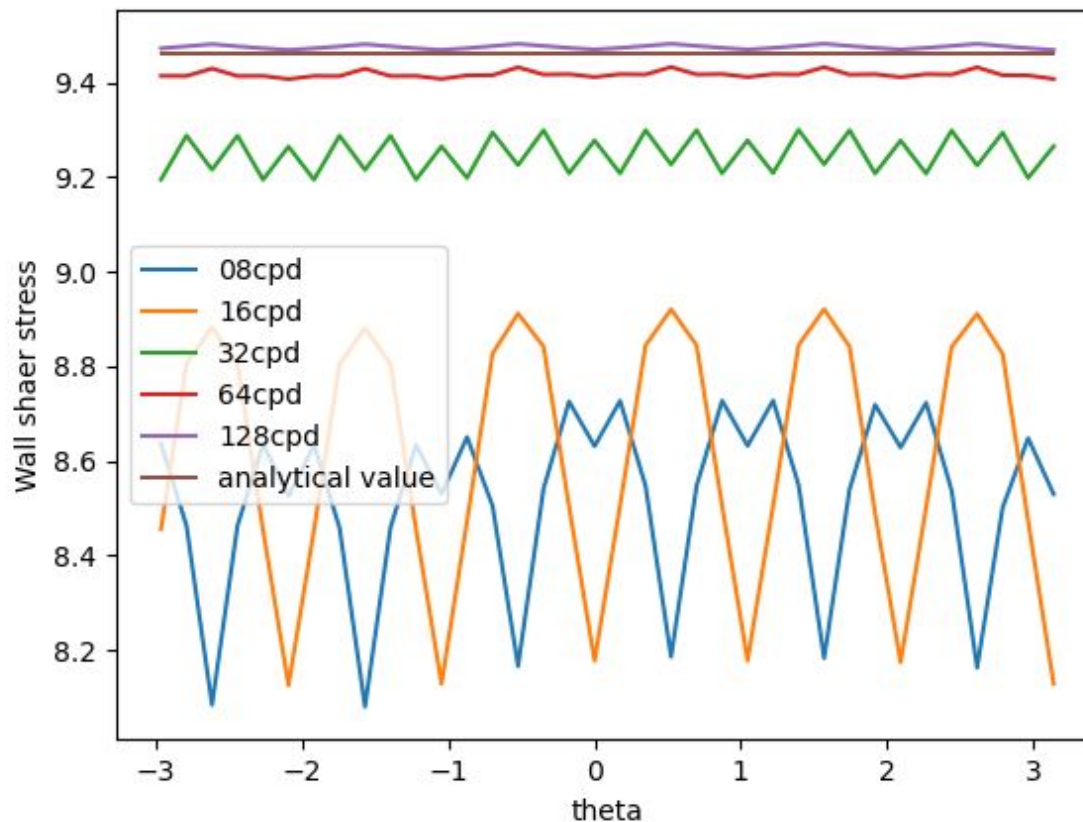
Orders of convergence cut cell: 0.94 1.42 1.02



- 1) Orders of convergence are based on three subsequent solutions.
- 2) For plotting it was assumed that the solution for the smallest dt is the exact solution.
- 3) u_{axial} at smallest dt :
 - a) analytical solution = 0.92333
 - b) ghost cell = 0.92329
 - c) cut cell = 0.92352

Wall shear stress in ghost cell

Calculation is based on the point 0.05 from the wall.



WSS is calculated by:

- 1) $\tau_w = \mu * du/dr$
- 2) correcting τ_w by the difference of analytical τ_w and τ at the $\frac{1}{2} dr$:

$$\text{correction} = (\tau_w - \tau(\frac{1}{2} dr))$$

Wall shear stress in cut cell

Wall shear stress is extracted from the solution for grids 8cpd...128cpd:

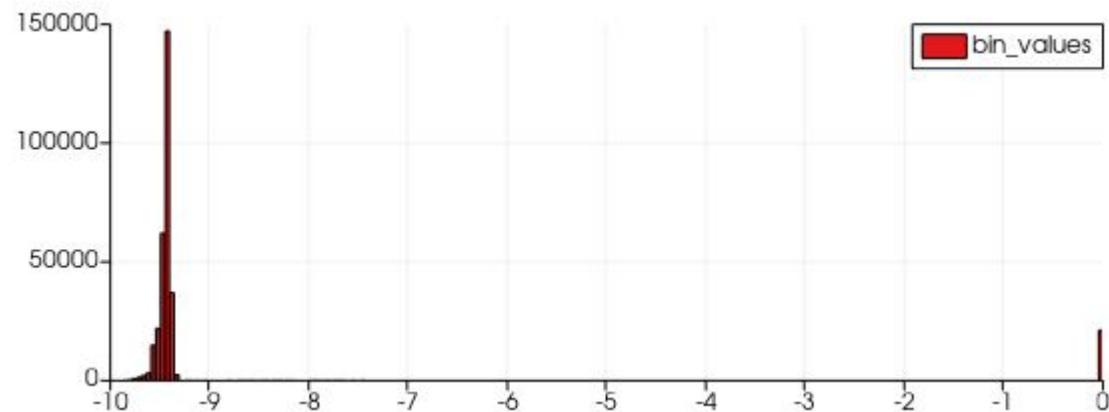
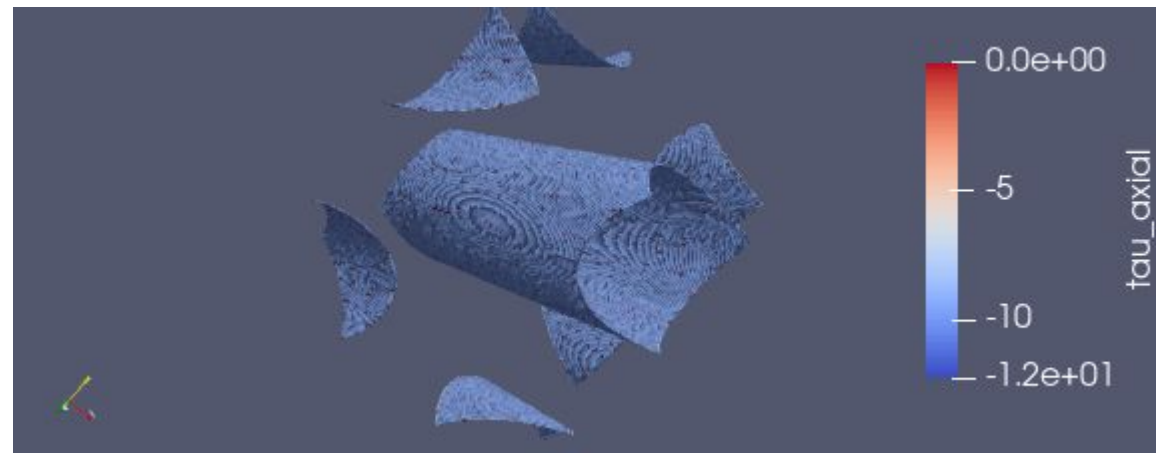
- 4.45 Pa
- 7.14 Pa
- 8.41 Pa
- 8.17 Pa
- 8.79 Pa

Exact analytical value is:

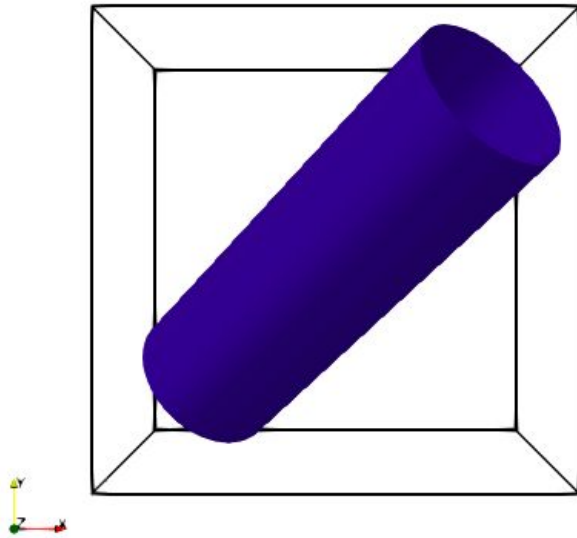
- 9.46 Pa

Peak of the histogram is at: 9.42 Pa

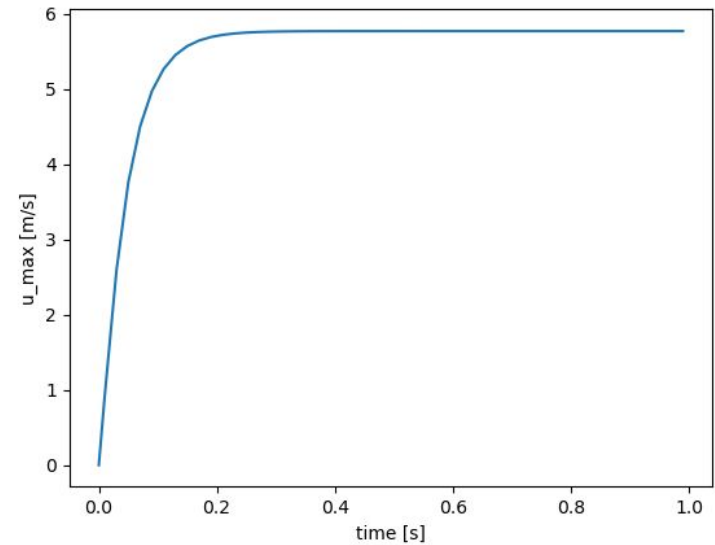
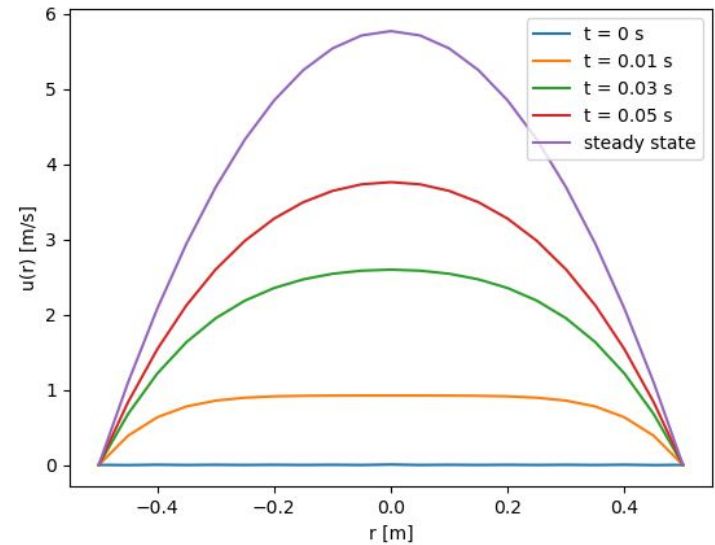
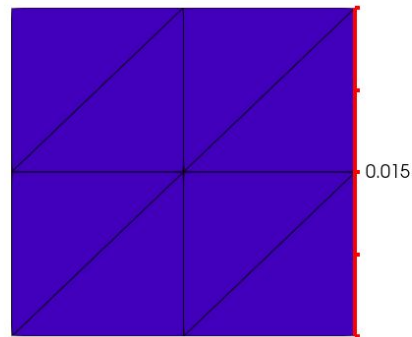
At some points **WSS=0**.



Oblique pipe flow



Pipe axis: $[1,1,1]$,
 diameter = 1
 GRADPX=-160
 RHO=1, GMOL=1,
 STL triangles
 resolution: $7.5E-3$



Grid study

Time stepping:

- $DT = 1E-5$
- $MTSTEP = 1E5$
- Total time of the simulation = 0.01 s

Pressure equation settings:

- $EPCORR = 1E-6$
- $MPCORR = 32$
- $MPCNACH = 4$
- Total maximal number of pressure iterations = 128

Grids:

- 8 cells per diameter, $dx = 0.125$
- 16 cells per diameter, $dx = 6.25E-2$
- 32 cells per diameter, $dx = 3.125E-2$
- 64 cells per diameter, $dx = 1.5625E-2$
- 128 cells per diameter, $dx = 7.1825E-3$

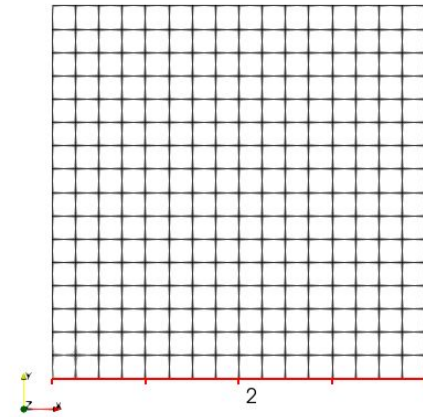


Fig 1: XY view of 08cpd grid

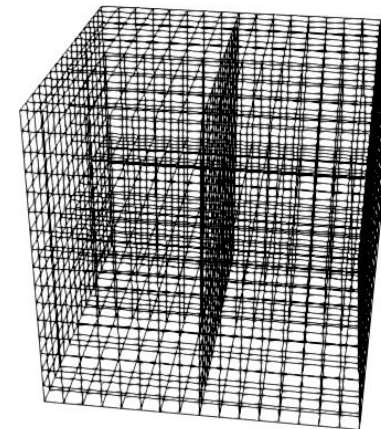


Fig 2: 3D view of 08cpd grid

Comparison with literature (time steps 10k-13k)

	Avg(Cd)	Cd'	Cl'	St
Uhlmann [2]	1.501	± 0.011	± 0.349	0.172
Uhlmann [2] enlarged domain	1.453	± 0.011	± 0.339	0.169
Liu et al. [3]	1.35	± 0.012	± 0.339	0.165
Present	1.4502	0.0086	0.342	0.1667